DGMS JSO GEOLOGY 2019 • GEOCHEMISTRY & COSMOCHEMISTRY

Question 94: Stellar Nucleosynthesis Beyond ^{209}Bi

Detailed Solution, Physical Bottlenecks, and Comparison of Stellar Nucleosynthetic Pathways

Q.94: Nuclides heavier than ^{209}Bi are produced by which of the following nucleosynthetic processes?

(a) Slow neutron capture (s-process) and p-p chain

(b) Rapid neutron capture (r-process) and CNO cycle

✓ **(c) Rapid neutron capture (r-process) and rapid proton capture (p-process)**

(d) Spallation of C and O nuclides (x-process)

Correct Option: (c) Rapid neutron capture (r-process) and rapid proton capture (p-process)

1. Cosmochemical Foundation & The Termination of the s-Process

Elements heavier than iron ($Z > 26$, $A > 56$) cannot be formed by exothermic thermonuclear fusion (such as H-burning, He-burning, C-burning, or Si-burning) because the nuclear binding energy per nucleon reaches its maximum at ^{56}Fe

and ^{62}Ni. Further element synthesis requires endothermic neutron-capture and proton-capture reactions:

- **Slow Neutron Capture (s-process):** The neutron capture rate is much slower than the beta-decay rate of unstable daughter nuclides ($\tau_{cap} \gg \tau_{\beta}$). It occurs primarily in the asymptotic giant branch (AGB) stars during thermal pulses.
- **Rapid Neutron Capture (r-process):** The neutron flux is extremely dense ($n_n > 10^{20}$ cm^{-3}), such that neutron captures occur far faster than beta decay ($\tau_{cap} \ll \tau_{\beta}$). It occurs in cataclysmic explosive environments such as core-collapse supernovae (Type II) and binary neutron star mergers (kilonovae).
- **p-process (Proton Capture / Photodisintegration):** Synthesizes rare, neutron-deficient (proton-rich) nuclides that cannot be produced by neutron capture because they are shielded behind stable isobars.

The Lead-Bismuth-Polonium s-Process Termination Cycle:

$$^{209}\text{Bi} + n \rightarrow {}^{210}\text{Bi} + \gamma$$

$$^{210}\text{Bi} \xrightarrow[t_{1/2}=5.01\text{ d}]{\beta^-} {}^{210}\text{Po} + e^- + \bar{\nu}_e$$

$$^{210}\text{Po} \xrightarrow[t_{1/2}=138.4\text{ d}]{\alpha} {}^{206}\text{Pb} + {}^{4}\text{He}$$ (Recycles back!)

Because ^{210}Po alpha-decays back to ^{206}Pb rather than capturing another neutron, the s-process forms a closed loop and **strictly terminates at ^{209}Bi**. It is physically impossible for the s-process to produce actinide elements such as Thorium (^{232}Th) and Uranium (^{235}U, ^{238}U).

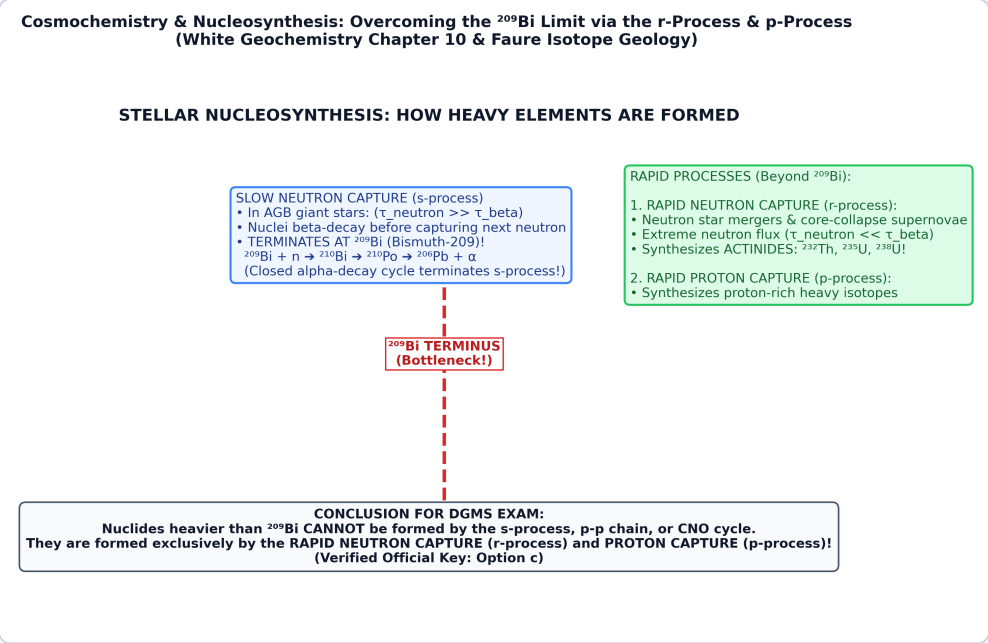


Figure 94.1: Chart of nuclides illustrating the termination of the s-process at ^{209}Bi via alpha-decay recycling of ^{210}Po, and the breakthrough to actinides via the rapid neutron capture (r-process). Based on White, Geochemistry (Chapter 10) and Faure, Principles of Isotope Geology.

2. Production of Trans-Bismuth Nuclides (Actinides) via the r-Process and p-Process

To cross the unstable alpha-decay valley beyond bismuth and synthesize long-lived actinides like ^{232}Th (half-life 14.05 Gyr), ^{238}U (4.47 Gyr), and ^{235}U (0.704 Gyr), a tremendously high neutron flux is required. Under r-process conditions:

- Neutrons are captured in fractions of a millisecond, driving the nucleus far to the neutron-rich side of the valley of beta-stability towards the neutron drip line before unstable nuclides have time to alpha-decay.
- Once the explosive neutron flux freezes out, these extremely neutron-rich isotopes cascade back to stability via successive β^- decays, populating the heavy elements all the way to uranium, thorium, and beyond.
- Concurrently, explosive nucleosynthesis produces heavy proton-rich nuclei via the **p-process** (consisting of rapid proton capture (p, γ) or photodisintegration (γ, n), (γ, p), and (γ, α) during supernova shock wave propagation through O-Ne-rich stellar mantles).

3. Systematic Option Evaluation

Option	Nucleosynthetic Processes Listed	Physical Evaluation	Validity
(a)	Slow neutron capture (s-process) and p-p chain	**Incorrect.** The p-p chain only operates in low-mass main-sequence stars (like our Sun) fusing Hydrogen into Helium ($^{1}H \rightarrow {}^{4}He$). The s-process strictly terminates at ^{209}Bi.	Invalid
(b)	Rapid neutron capture (r-process) and CNO cycle	**Incorrect.** While the r-process produces nuclides beyond ^{209}Bi, the CNO cycle is a catalytic hydrogen-burning cycle in intermediate-to-high mass stars that only cycles C, N, and O isotopes.	Invalid
(c)	**Rapid neutron capture (r-process) and rapid proton capture (p-process)**	**Correct.** The r-process is the primary mechanism bypassing the alpha-decay bottleneck at ^{210}Po to build actinides; the p-process synthesizes trans-iron and heavy proton-rich nuclides in stellar explosions.	**Correct**
(d)	Spallation of C and O nuclides (x-process)	**Incorrect.** Cosmic ray spallation (the x-process) fragments heavier target nuclei (like C, N, O) into light, fragile elements: Lithium (Li), Beryllium (Be), and Boron (B). It cannot produce heavy elements.	Invalid

Key Geochemical Takeaway

^{209}Bi is the heaviest stable nuclide in the universe and the ultimate ceiling for the slow neutron capture process (s-process). All primordial actinides found on Earth today (^{232}Th, ^{235}U, ^{238}U), which power Earth's internal radioactive heat engine, were forged exclusively through the **r-process** during neutron star mergers and core-collapse supernovae prior to the collapse of the solar nebula.

4. Standard Textbook References

- **William M. White**, *Geochemistry* (2020), Chapter 10: "Cosmochemistry and Nucleosynthesis", Section 10.2: Nucleosynthesis Beyond the Iron Peak, pp. 415–430.
- **Gunter Faure & Teresa M. Mensing**, *Isotopes: Principles and Applications* (3rd Edition, 2005), Chapter 1: "Origin of the Chemical Elements", pp. 1–22.
- **E. M. Burbidge, G. R. Burbidge, W. A. Fowler, & F. Hoyle (B^{2}FH)**, *Synthesis of the Elements in Stars*, Reviews of Modern Physics, 29(4), pp. 547–650 (1957).